Oracle ODS

Ambiente criado para suportar a geração de relatórios no Sabre, as bases SABREMC, SABRECC e SABRELM foram replicadas utilizando o Data Guard, no modo Logical Standby, que permite utilizar o banco de dados no modo somente leitura. Criado também a base replicada do Cargo, CARGO_ODS.

Sistema Operacional

- Red Hat Enterprise Linux Server release 5.8 (Tikanga)

Arquitetura

- Binários em uma LUN de 10Gb formatada como ext3 e montada no filesystem /u01 (Oracle Restart / Listener)
- Binários e dados alocados em uma LUN de 150Gb formatada como ext3 e montada no filesystem /u02 (SABRE)
- Binários e dados alocados em uma LUN de 100Gb formatada como ext3 e montada no filesystem /u03 (CARGO)

VMWare

NFS

Para realizar backups via RMAN ou export (Data Pump) em NFS, montar da seguinte forma:

```
-t nfs -o rsize=65536,wsize=65536,hard,intr,nodev,nosuid,actimeo=0,nfsvers=3
```

Atualmente existe o filesystem /work-tmp que é montado automaticamente no reboot e já esta utilizando estes parametros.

Dependencias de SO

Instale as dependências do RHEL 5 com os comandos abaixo:

```
yum groupinstall 'Development Tools' 'Development Libraries'
yum install libXp compat-db control-center compat-gcc libstdc++ libaio sysstat
yum install compat-libstdc++-33
yum install libaio-devel
yum install libXtst
```

Filesystem

```
/dev/sda1               /                       ext3    defaults                        1 1
tmpfs                   /dev/shm                tmpfs   defaults                        0 0
devpts                  /dev/pts                devpts  gid=5,mode=620                  0 0
sysfs                   /sys                    sysfs   defaults                        0 0
proc                    /proc                   proc    defaults                        0 0
# Swap
/dev/vg_data/swap0      swap                    swap    defaults                        0 0
#
/dev/vg/var             /var                    ext3    defaults                        1 2
/dev/vg/home            /home                   ext3    defaults                        1 2
```

```
/dev/vg/tmp             /tmp                    ext3    defaults                        1 2
/dev/vg_data01/U01      /u01                    ext3    defaults                        1 2
/dev/vg_data02/U02      /u02                    ext3    defaults                        1 2
/dev/vg_data03/U03      /u03                    ext3    defaults                        1 2
# NFS
10.0.6.210:/home/work-tmp     /work-tmp               nfs  rsize=65536,wsize=65536,hard,intr,noc
```

Endereços IP

ORAODSTB578A

- IP Físico: 10.0.6.50 (ORAODSTB578A.azul.corp)
- IP Diveo: 192.168.254.42

Instâncias

As seguintes instâncias estão configuradas neste ambiente:

Sistema	Instâncias	Porta	Home	Device	User Linux	Migrado	Versão	CPU Patch	DBA AZUL
Oracle Restart	-	1521	/u01	/dev/mapper /vg_data01-U01	oragrid	SIM	11.2.0.4	-	-
Sabre MC Report	SBMC_ODS	1521	/u02	/dev/mapper /vg_data02-U02	sabre	SIM	10.2.0.4	11725015	SIM
Sabre CC Report	SBCC_ODS	1521	/u02	/dev/mapper /vg_data02-U02	sabre	SIM	10.2.0.4	11725015	SIM
Sabre LM Report	SBLM_ODS	1521	/u02	/dev/mapper /vg_data02-U02	sabre	SIM	10.2.0.4	11725015	SIM
CARGO DB Report	CARGO_ODS	1521	/u03	/dev/mapper /vg_data03-U03	cargo	SIM	10.2.0.4	11725015	SIM

TNSNAMES

CARGO_ODS

```
CARGO_ODS =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = ORAODSTB578A)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = CARGO_ODS)
    )
  )
```

SBMC_ODS

```
SBMC_ODS =
  (DESCRIPTION =
```

```
    (ADDRESS = (PROTOCOL = TCP)(HOST = ORAODSTB578A)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = SBMC_ODS)
    )
  )
```

SBCC_ODS

```
SBCC_ODS =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = ORAODSTB578A)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = SBCC_ODS)
    )
  )
```

SBLM_ODS

```
SBLM_ODS =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = ORAODSTB578A)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = SBLM_ODS)
    )
  )
```

Configuração dos DBLinks

Na configuração do ambiente ODS, as bases do Sabre estão interligadas por DBLink, todos os acessos devem ser realizados através do banco SBCC_ODS, abaixo seguem os comandos utilizados na configuração dos ambientes, a liberação do acesso deve ser realizado na base de origem e depois criado um sinonimo na base SBCC_ODS.

SBMC

```
CREATE USER ods_cc IDENTIFIED BY azul2010;
GRANT CONNECT TO ods_cc;
GRANT SELECT ON azumcdba.ftdaily TO ods_cc;
GRANT SELECT ON AZUMCDBA.ctffout_crewmember TO ods_cc;
```

SBLM

```
CREATE USER ods_cc IDENTIFIED BY azul2010;
GRANT CONNECT TO ods_cc;
```

SBCC

```
CREATE PUBLIC DATABASE LINK ODS_MC CONNECT TO ods_cc IDENTIFIED BY azul2010 USING 'SBMC_ODS';
CREATE PUBLIC DATABASE LINK ODS_LM CONNECT TO ods_cc IDENTIFIED BY azul2010 USING 'SBLM_ODS';
CREATE PUBLIC SYNONYM FTDAILY FOR AZUMCDBA.FTDAILY@ODS_MC;
CREATE PUBLIC SYNONYM CTFFOUT_CREWMEMBER FOR AZUMCDBA.CTFFOUT_CREWMEMBER@ODS_MC;
```

Configuração

SABRE CC

- Desabilitar a configuração do Data Guard Broker:

```
dgmgrl /
DGMGRL> DISABLE CONFIGURATION;
DGMGRL> exit
```

- Habilitar o FORCE LOGGING no banco de dados de produção

```
SQL> alter database force logging;
```

- Backup do banco de dados de produção - SABRECC

```
run{
 allocate channel ch00 device type disk;
 allocate channel ch01 device type disk;
 allocate channel ch02 device type disk;
 allocate channel ch03 device type disk;
 backup format '/work-tmp/sabre/SABRECC_DB_%U.bkp' database;
 sql 'alter system archive log current';
 backup format '/work-tmp/sabre/SABRECC_ARCH_%U.bkp' archivelog from time 'SYSDATE-12/24';
 backup format '/work-tmp/sabre/controlstdSABRECC.bkp' current controlfile for standby;
}
```

- Restore do banco de dados no ambiente standby

```
run{
 restore standby controlfile from '/work-tmp/sabre/controlstdSABRECC.bkp';
 sql 'alter database mount';
 set newname for tempfile 1 to '/u02/app/oracle/oradata/SBCC_ODS/tstemp01.dbf';
 set newname for tempfile 2 to '/u02/app/oracle/oradata/SBCC_ODS/tstemp02.dbf';
 set newname for tempfile 5 to '/u02/app/oracle/oradata/SBCC_ODS/tstemp05.dbf';
 switch tempfile all;
 set newname for datafile 1 to '/u02/app/oracle/oradata/SBCC_ODS/system01.dbf';
 set newname for datafile 2 to '/u02/app/oracle/oradata/SBCC_ODS/undotbs01.dbf';
 set newname for datafile 3 to '/u02/app/oracle/oradata/SBCC_ODS/sysaux01.dbf';
 set newname for datafile 4 to '/u02/app/oracle/oradata/SBCC_ODS/users01.dbf';
 set newname for datafile 5 to '/u02/app/oracle/oradata/SBCC_ODS/undotbs02.dbf';
 set newname for datafile 6 to '/u02/app/oracle/oradata/SBCC_ODS/sabre_data01.dbf';
 set newname for datafile 7 to '/u02/app/oracle/oradata/SBCC_ODS/sabre_data02.dbf';
 set newname for datafile 8 to '/u02/app/oracle/oradata/SBCC_ODS/sabre_data03.dbf';
 allocate channel ch00 device type disk;
 allocate channel ch01 device type disk;
 allocate channel ch02 device type disk;
 allocate channel ch03 device type disk;
 restore database;
 switch datafile all;
 recover database;
}
```

- Criar os standby REDOLOGs

```
SQL> ALTER DATABASE ADD STANDBY LOGFILE THREAD 1 GROUP 9 SIZE 52428800, GROUP 10 SIZE 52428800,
     GROUP 11 SIZE 52428800, GROUP 12 SIZE 52428800, GROUP 13 SIZE 52428800;
SQL> ALTER DATABASE ADD STANDBY LOGFILE THREAD 2 GROUP 14 SIZE 52428800, GROUP 15 SIZE 52428800,
     GROUP 16 SIZE 52428800, GROUP 17 SIZE 52428800, GROUP 18 SIZE 52428800;
```

- Alterar os parametros no banco primary (SABRECC)

```
alter system set LOG_ARCHIVE_CONFIG='DG_CONFIG=(SABRECC,SBCC_ODS)' scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_1='LOCATION=+DGARCH VALID_FOR=(ALL_LOGFILES,ALL_ROLES)
     DB_UNIQUE_NAME=SABRECC' scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_2='SERVICE=SBCC_ODS LGWR ASYNC AFFIRM REOPEN=15 NET_TIMEOUT=15
     VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=SBCC_ODS' scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_STATE_1=ENABLE scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_STATE_2=ENABLE scope=both sid='*';
alter system set LOG_ARCHIVE_MAX_PROCESSES=30 scope=both sid='*';
alter system set FAL_SERVER=SBCC_ODS scope=both sid='*';
alter system set FAL_CLIENT=SABRECC scope=both sid='*';
```

- Alterar no banco Logical Standby

```
alter system set LOG_ARCHIVE_CONFIG='DG_CONFIG=(SABRECC,SBCC_ODS)' scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_1='LOCATION=/u02/app/oracle/oraarch/SBCC_ODS/
      VALID_FOR=(ALL_LOGFILES,ALL_ROLES) DB_UNIQUE_NAME=SBCC_ODS';
alter system set LOG_ARCHIVE_DEST_2='SERVICE=SABRECC LGWR ASYNC
      VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=SABRECC';
alter system set LOG_ARCHIVE_DEST_STATE_1=ENABLE scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_STATE_2=ENABLE scope=both sid='*';
alter system set LOG_ARCHIVE_MAX_PROCESSES=2 scope=both sid='*';
alter system set STANDBY_FILE_MANAGEMENT=AUTO scope=both sid='*';
alter system set FAL_SERVER=SABRECC scope=both sid='*';
alter system set FAL_CLIENT=SBCC_ODS scope=both sid='*';
```

- Habiliar o Physical standby:

```
SQL> ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DISCONNECT FROM SESSION;
```

- Monitorando a transferencia dos blocos do ambiente de produção para o ambiente Standby:

```
select PROCESS, STATUS, SEQUENCE#, THREAD#, BLOCK#, BLOCKS,
     TO_CHAR(SYSDATE,'DD-MON-YYYY HH24:MI:SS') TIME from V$MANAGED_STANDBY;
```

- Parar a replicação no Standby para habilitar a replicação Logical:

```
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;
```

- Alterar os parametro no banco principal(SABRECC):

```
alter system set LOG_ARCHIVE_DEST_1='LOCATION=+DGARCH VALID_FOR=(ONLINE_LOGFILES,ALL_ROLES)
          DB_UNIQUE_NAME=SABRECC' scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_3='LOCATION=+DGDATA02 VALID_FOR=(STANDBY_LOGFILES,STANDBY_ROLE
          DB_UNIQUE_NAME=SABRECC' scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_STATE_3=ENABLE scope=both sid='*';
```

- Criando o dicionario de dados do Logical Standby - Base principal, este comando é executado na base principal:

```
EXECUTE DBMS_LOGSTDBY.BUILD;
```

- Convertendo a base de dados phisical para logical:

```
ALTER DATABASE RECOVER TO LOGICAL STANDBY SBCC_ODS;
SHUTDOWN IMMEDIATE;
STARTUP MOUNT;
```

- Alterar os parametros no base standby:

```
alter system set LOG_ARCHIVE_DEST_1='LOCATION=/u02/app/oracle/oraarch/SBCC_ODS/
      VALID_FOR=(ONLINE_LOGFILES,ALL_ROLES) DB_UNIQUE_NAME=SBCC_ODS';
alter system set LOG_ARCHIVE_DEST_2='SERVICE=SABRECC LGWR ASYNC
      VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=SABRECC';
alter system set LOG_ARCHIVE_DEST_3='LOCATION=/u02/app/oracle/standby/SBCC_ODS/
      VALID_FOR=(STANDBY_LOGFILES,STANDBY_ROLE) DB_UNIQUE_NAME=SBCC_ODS' scope=both sid='*';
```

- Abrindo o banco logical standby:

```
ALTER DATABASE OPEN RESETLOGS;
```

- Iniciando o Logical Apply no Standby:

```
ALTER DATABASE START LOGICAL STANDBY APPLY IMMEDIATE;
```

- Habilitar a configuração do Data Guard Broker:

```
dgmgrl /
DGMGRL> ENABLE CONFIGURATION;
DGMGRL> exit
```

SABRE MC

- Desabilitar a configuração do Data Guard Broker:

```
dgmgrl /
DGMGRL> DISABLE CONFIGURATION;
DGMGRL> exit
```

* Habilitar o FORCE LOGGING no banco de dados de produção

```
SQL> alter database force logging;
```

- Backup do banco de dados de produção - SABREMC

```
run{
 allocate channel ch00 device type disk;
 allocate channel ch01 device type disk;
 allocate channel ch02 device type disk;
 allocate channel ch03 device type disk;
 backup format '/work-tmp/sabre/SABREMC_DB_%U.bkp' database;
 sql 'alter system archive log current';
 backup format '/work-tmp/sabre/SABREMC_ARCH_%U.bkp' archivelog from time 'SYSDATE-12/24';
 backup format '/work-tmp/sabre/controlSABREMC.bkp' current controlfile for standby;
}
```

- Restore do banco de dados no ambiente standby

```
run{
 restore standby controlfile from '/work-tmp/sabre/controlSABREMC.bkp';
 sql 'alter database mount';
 set newname for tempfile 1 to '/u02/app/oracle/oradata/SBMC_ODS/tstemp01.dbf';
 switch tempfile all;
 set newname for datafile 1 to '/u02/app/oracle/oradata/SBMC_ODS/system01.dbf';
 set newname for datafile 2 to '/u02/app/oracle/oradata/SBMC_ODS/undotbs01.dbf';
 set newname for datafile 3 to '/u02/app/oracle/oradata/SBMC_ODS/sysaux01.dbf';
 set newname for datafile 4 to '/u02/app/oracle/oradata/SBMC_ODS/users01.dbf';
 set newname for datafile 5 to '/u02/app/oracle/oradata/SBMC_ODS/undotbs02.dbf';
 set newname for datafile 6 to '/u02/app/oracle/oradata/SBMC_ODS/sabre_data.dbf';
 allocate channel ch00 device type disk;
 allocate channel ch01 device type disk;
 allocate channel ch02 device type disk;
 allocate channel ch03 device type disk;
 restore database;
 switch datafile all;
 recover database;
}
```

- Criar os standby REDOLOGs

```
ALTER DATABASE ADD STANDBY LOGFILE THREAD 1 GROUP 9 SIZE 134217728, GROUP 10 SIZE 134217728,
        GROUP 11 SIZE 134217728, GROUP 12 SIZE 134217728, GROUP 13 SIZE 134217728;
ALTER DATABASE ADD STANDBY LOGFILE THREAD 2 GROUP 14 SIZE 134217728, GROUP 15 SIZE 134217728,
        GROUP 16 SIZE 134217728, GROUP 17 SIZE 134217728, GROUP 18 SIZE 134217728;
```

- Alterar no banco Logical Standby

```
alter system set LOG_ARCHIVE_CONFIG='DG_CONFIG=(SABREMC,SBMC_ODS)' scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_1='LOCATION=/u02/app/oracle/oraarch/SBMC_ODS/
      VALID_FOR=(ALL_LOGFILES,ALL_ROLES) DB_UNIQUE_NAME=SBMC_ODS';
alter system set LOG_ARCHIVE_DEST_2='SERVICE=SABREMC LGWR ASYNC
      VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=SABREMC';
alter system set LOG_ARCHIVE_DEST_STATE_1=ENABLE scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_STATE_2=ENABLE scope=both sid='*';
alter system set LOG_ARCHIVE_MAX_PROCESSES=2 scope=both sid='*';
alter system set STANDBY_FILE_MANAGEMENT=AUTO scope=both sid='*';
alter system set FAL_SERVER=SABREMC scope=both sid='*';
alter system set FAL_CLIENT=SBCM_ODS scope=both sid='*';
alter system set DB_FILE_NAME_CONVERT='+DGDATA02/sabremc/datafile/','/u02/app/oracle/oradata/SBM
        '+DGDATA02/sabrecc/tempfile/','/u02/app/oracle/oradata/SBMC_ODS/' scope=spfile sid='*';
alter system set STANDBY_FILE_MANAGEMENT=AUTO scope=both sid='*';
```

- Habiliar o standby:

```
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DISCONNECT FROM SESSION;
```

- Alterar os parametros no banco primary (SABREMC)

```
alter system set LOG_ARCHIVE_CONFIG='DG_CONFIG=(SABREMC,SBMC_ODS)' scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_1='LOCATION=+DGARCH VALID_FOR=(ALL_LOGFILES,ALL_ROLES)
    DB_UNIQUE_NAME=SABREMC' scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_2='SERVICE=SBMC_ODS LGWR ASYNC AFFIRM REOPEN=15 NET_TIMEOUT=15
    VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=SBMC_ODS' scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_STATE_1=ENABLE scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_STATE_2=ENABLE scope=both sid='*';
```

```
alter system set LOG_ARCHIVE_MAX_PROCESSES=30 scope=both sid='*';
alter system set FAL_SERVER=SBCM_ODS scope=both sid='*';
alter system set FAL_CLIENT=SABRECM scope=both sid='*';
```

- Monitorar:

```
select PROCESS, STATUS, SEQUENCE#, THREAD#, BLOCK#, BLOCKS,
      TO_CHAR(SYSDATE,'DD-MON-YYYY HH24:MI:SS') TIME from V$MANAGED_STANDBY
```

- Parar a replicacao no Standby:

```
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;
```

- Alterar o parametro no banco principal

```
alter system set LOG_ARCHIVE_DEST_1='LOCATION=+DGARCH
    VALID_FOR=(ONLINE_LOGFILES,ALL_ROLES) DB_UNIQUE_NAME=SABREMC' scope=both sid='*';
```

- Criando o dicionario de dados do Logical Standby - Base principal:

```
EXECUTE DBMS_LOGSTDBY.BUILD;
```

- Convertendo a base de dados phisical para logical:

```
ALTER DATABASE RECOVER TO LOGICAL STANDBY SBMC_ODS;
SHUTDOWN IMMEDIATE;
STARTUP MOUNT;
```

- Alterando os parametros no base standby

```
alter system set LOG_ARCHIVE_DEST_1='LOCATION=/u02/app/oracle/oraarch/SBCC_ODS/
      VALID_FOR=(ONLINE_LOGFILES,ALL_ROLES) DB_UNIQUE_NAME=SBMC_ODS' scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_2='SERVICE=SABREMC LGWR ASYNC
      VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=SABREMC'  scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_3='LOCATION=/u02/app/oracle/standby/SBMC_ODS/
      VALID_FOR=(STANDBY_LOGFILES,STANDBY_ROLE) DB_UNIQUE_NAME=SBMC_ODS' scope=both sid='*';
```

- Abrindo o banco logical standby:

```
ALTER DATABASE OPEN RESETLOGS;
```

- Iniciando o Logical Apply no Standby:

```
ALTER DATABASE START LOGICAL STANDBY APPLY IMMEDIATE;
```

- Habilitar a configuração do Data Guard Broker:

```
dgmgrl /
DGMGRL> ENABLE CONFIGURATION;
DGMGRL> exit
```

- Depois que a replicação estiver operacional, realizar a configuração abaixo para remover a tabela FTSEND da replização:

```
ALTER DATABASE STOP LOGICAL STANDBY APPLY;
EXECUTE DBMS_LOGSTDBY.SKIP (stmt => 'DML', schema_name => 'AZUMCDBA',
     object_name => 'FTSEND', proc_name => NULL);
ALTER DATABASE START LOGICAL STANDBY APPLY IMMEDIATE;
```

SABRE LM

- Desabilitar a configuração do Data Guard Broker:

```
dgmgrl /
DGMGRL> DISABLE CONFIGURATION;
DGMGRL> exit
```

- Habilitar o FORCE LOGGING no banco de dados de produção

```
SQL> alter database force logging;
```

- Backup do banco de dados de produção - SABRELM

```
run{
 allocate channel ch00 device type disk;
 allocate channel ch01 device type disk;
 allocate channel ch02 device type disk;
 allocate channel ch03 device type disk;
 backup format '/work-tmp/sabre/SABRELM_DB_%U.bkp' database;
 sql 'alter system archive log current';
 backup format '/work-tmp/sabre/SABRELM_ARCH_%U.bkp' archivelog from time 'SYSDATE-12/24';
 backup format '/work-tmp/sabre/controlstdSABRELM.bkp' current controlfile for standby;
}
```

- Restore do banco de dados no ambiente standby

```
run{
 restore standby controlfile from '/work-tmp/sabre/controlstdSABRELM.bkp';
 sql 'alter database mount';
 set newname for tempfile 1 to '/u02/app/oracle/oradata/SBLM_ODS/tstemp01.dbf';
 switch tempfile all;
 set newname for datafile 1 to '/u02/app/oracle/oradata/SBLM_ODS/system01.dbf';
 set newname for datafile 2 to '/u02/app/oracle/oradata/SBLM_ODS/undotbs01.dbf';
 set newname for datafile 3 to '/u02/app/oracle/oradata/SBLM_ODS/sysaux01.dbf';
 set newname for datafile 4 to '/u02/app/oracle/oradata/SBLM_ODS/users01.dbf';
 set newname for datafile 5 to '/u02/app/oracle/oradata/SBLM_ODS/undotbs02.dbf';
 allocate channel ch00 device type disk;
 allocate channel ch01 device type disk;
 allocate channel ch02 device type disk;
 allocate channel ch03 device type disk;
 restore database;
 switch datafile all;
 recover database;
}
```

- Criar os standby REDOLOGs

```
SQL> ALTER DATABASE ADD STANDBY LOGFILE THREAD 1 GROUP 9 SIZE 52428800, GROUP 10 SIZE 52428800,
```

```
    GROUP 11 SIZE 52428800, GROUP 12 SIZE 52428800, GROUP 13 SIZE 52428800;
SQL> ALTER DATABASE ADD STANDBY LOGFILE THREAD 2 GROUP 14 SIZE 52428800, GROUP 15 SIZE 52428800,
    GROUP 16 SIZE 52428800, GROUP 17 SIZE 52428800, GROUP 18 SIZE 52428800;
```

- Alterar os parametros no banco primary (SABRELM)

```
alter system set LOG_ARCHIVE_CONFIG='DG_CONFIG=(SABRELM,SBLM_ODS)' scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_1='LOCATION=+DGARCH VALID_FOR=(ALL_LOGFILES,ALL_ROLES)
    DB_UNIQUE_NAME=SABRELM' scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_2='SERVICE=SBLM_ODS LGWR ASYNC AFFIRM REOPEN=15 NET_TIMEOUT=15
    VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=SBLM_ODS' scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_STATE_1=ENABLE scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_STATE_2=ENABLE scope=both sid='*';
alter system set LOG_ARCHIVE_MAX_PROCESSES=30 scope=both sid='*';
alter system set FAL_SERVER=SBLM_ODS scope=both sid='*';
alter system set FAL_CLIENT=SABRELM scope=both sid='*';
```

- Alterar no banco Logical Standby

```
alter system set LOG_ARCHIVE_CONFIG='DG_CONFIG=(SABRELM,SBLM_ODS)' scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_1='LOCATION=/u02/app/oracle/oraarch/SBLM_ODS/
     VALID_FOR=(ALL_LOGFILES,ALL_ROLES) DB_UNIQUE_NAME=SBCC_ODS';
alter system set LOG_ARCHIVE_DEST_2='SERVICE=SABRELM LGWR ASYNC
     VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=SABRELM';
alter system set LOG_ARCHIVE_DEST_STATE_1=ENABLE scope=both sid='*';
alter system set LOG_ARCHIVE_DEST_STATE_2=ENABLE scope=both sid='*';
alter system set LOG_ARCHIVE_MAX_PROCESSES=2 scope=both sid='*';
alter system set STANDBY_FILE_MANAGEMENT=AUTO scope=both sid='*';
alter system set FAL_SERVER=SABRELM scope=both sid='*';
alter system set FAL_CLIENT=SBLM_ODS scope=both sid='*';
```

- Habiliar o Physical standby:

```
SQL> ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DISCONNECT FROM SESSION;
```

- Monitorando a transferencia dos blocos do ambiente de produção para o ambiente Standby:

```
select PROCESS, STATUS, SEQUENCE#, THREAD#, BLOCK#, BLOCKS,
     TO_CHAR(SYSDATE,'DD-MON-YYYY HH24:MI:SS') TIME from V$MANAGED_STANDBY
```

- Parar a replicação no Standby para habilitar a replicação Logical:

```
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;
```

- Alterar os parametro no banco principal(SABRELM):

```
alter system set LOG_ARCHIVE_DEST_1='LOCATION=+DGARCH
  VALID_FOR=(ONLINE_LOGFILES,ALL_ROLES) DB_UNIQUE_NAME=SABRELM' scope=both sid='*';
```

- Criando o dicionario de dados do Logical Standby - Base principal, este comando é executado na base principal:

```
EXECUTE DBMS_LOGSTDBY.BUILD;
```

- Convertendo a base de dados phisical para logical:

```
ALTER DATABASE RECOVER TO LOGICAL STANDBY SBLM_ODS;
SHUTDOWN IMMEDIATE;
STARTUP MOUNT;
```

- Alterar os parametros no base standby:

```
alter system set LOG_ARCHIVE_DEST_1='LOCATION=/u02/app/oracle/oraarch/SBLM_ODS/
      VALID_FOR=(ONLINE_LOGFILES,ALL_ROLES) DB_UNIQUE_NAME=SBLM_ODS';
alter system set LOG_ARCHIVE_DEST_2='SERVICE=SABRELM LGWR ASYNC
      VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=SABRELM';
alter system set LOG_ARCHIVE_DEST_3='LOCATION=/u02/app/oracle/standby/SBLM_ODS/
      VALID_FOR=(STANDBY_LOGFILES,STANDBY_ROLE) DB_UNIQUE_NAME=SBLM_ODS' scope=both sid='*';
```

- Abrindo o banco logical standby:

```
ALTER DATABASE OPEN RESETLOGS;
```

- Iniciando o Logical Apply no Standby:

```
ALTER DATABASE START LOGICAL STANDBY APPLY IMMEDIATE;
```

- Habilitar a configuração do Data Guard Broker:

```
dgmgrl /
DGMGRL> ENABLE CONFIGURATION;
DGMGRL> exit
```

CARGODB ODS

Produção: ORAFDBTB578A / ORAFDBTB578578B - Instância: CARGODB

- Desabilitar a configuração do Data Guard Broker:

```
dgmgrl /
DGMGRL> DISABLE CONFIGURATION;
DGMGRL> exit
```

- Habilitar o FORCE LOGGING no banco de dados de produção

```
SQL> alter database force logging;
```

- Backup PROD

```
su - cargo
rman target /
run{
 allocate channel ch00 device type disk;
 allocate channel ch01 device type disk;
 allocate channel ch02 device type disk;
 allocate channel ch03 device type disk;
```

```
backup format '/work-tmp/cargo/CARGODB_%U.bkp' database;
sql 'alter system archive log current';
backup format '/work-tmp/cargo/CARGODB_ARCH_%U.bkp' archivelog from time 'SYSDATE-12/24';
backup format '/work-tmp/cargo/controlCARGODB.bkp' current controlfile for standby;
}
```

- Restore CARGO ODS

```
run{
 restore standby controlfile from '/work-tmp/cargo/controlCARGODB.bkp';
 sql 'alter database mount';
}
```

- Como o CARGO de produção tem vários arquivos, é necessário renomear os arquivos, utilizar a saida do comando abaixo:

```
set linesize 120
set pagesize 0
set head off
select 'SET NEWNAME FOR DATAFILE '||FILE#||' TO NEW;' from v$datafile
```

- Pegar a saida o script e colocar no bloco abaixo, antes dos comandos allocate channel

```
run{
 <<colar saida aqui>>;
 allocate channel ch00 device type disk;
 allocate channel ch01 device type disk;
 allocate channel ch02 device type disk;
 allocate channel ch03 device type disk;
 restore database;
 switch datafile all;
 recover database delete archivelog;
}
```

- Criar os standby REDOLOGs

```
ALTER DATABASE ADD STANDBY LOGFILE THREAD 1 GROUP 9 SIZE 805306368, GROUP 10 SIZE 805306368,
        GROUP 11 SIZE 805306368, GROUP 12 SIZE 805306368, GROUP 13 SIZE 805306368;
ALTER DATABASE ADD STANDBY LOGFILE THREAD 2 GROUP 14 SIZE 805306368, GROUP 15 SIZE 805306368,
        GROUP 16 SIZE 805306368, GROUP 17 SIZE 805306368, GROUP 18 SIZE 805306368;
```

- Alterar os parametros no banco primary (CARGO)

```
alter system set LOG_ARCHIVE_DEST_3='SERVICE=CARGO_ODS LGWR ASYNC AFFIRM REOPEN=15
   NET_TIMEOUT=15 VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=CARGO_ODS' scope=both
alter system set STANDBY_FILE_MANAGEMENT=AUTO scope=both sid='*';
```

- Alterar no banco Logical Standby

```
alter system set LOG_ARCHIVE_DEST_1='LOCATION=/u01/app/oracle/oraarch/CARGO_ODS/
          VALID_FOR=(ALL_LOGFILES,ALL_ROLES) DB_UNIQUE_NAME=CARGO_ODS';
alter system set LOG_ARCHIVE_MAX_PROCESSES=2 scope=both sid='*';
alter system set STANDBY_FILE_MANAGEMENT=AUTO scope=both sid='*';
```

- Habiliar o standby:

```
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DISCONNECT FROM SESSION;
```

- Monitorar

```
select PROCESS, STATUS, SEQUENCE#, THREAD#, BLOCK#, BLOCKS,
      TO_CHAR(SYSDATE,'DD-MON-YYYY HH24:MI:SS') TIME from V$MANAGED_STANDBY
```

Verifica se as sequencias estão sendo aplicadas corretamente:

```
SELECT ARCH.THREAD# "Thread", ARCH.SEQUENCE# "Last Sequence Received", APPL.SEQUENCE# "Last Sequ
FROM
(SELECT THREAD# ,SEQUENCE# FROM V$ARCHIVED_LOG WHERE (THREAD#,FIRST_TIME )
    IN (SELECT THREAD#,MAX(FIRST_TIME) FROM V$ARCHIVED_LOG GROUP BY THREAD#)) ARCH,
(SELECT THREAD# ,SEQUENCE# FROM V$LOG_HISTORY WHERE (THREAD#,FIRST_TIME )
    IN (SELECT THREAD#,MAX(FIRST_TIME) FROM V$LOG_HISTORY GROUP BY THREAD#)) APPL
WHERE
ARCH.THREAD# = APPL.THREAD#
ORDER BY 1;
```

- Para a replicacao Standby:

Parar apenas a replicação se ela estiver sincronizada:

```
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;
```

- Criando o dicionario de dados do Logical Standby - Base principal(CARGO):

```
EXECUTE DBMS_LOGSTDBY.BUILD;
```

- Convertendo a base de dados physical para logical:

```
ALTER DATABASE RECOVER TO LOGICAL STANDBY CARGOODS;
```

- Abrindo o banco logical standby:

```

ALTER DATABASE OPEN RESETLOGS;
```

- Iniciando o Logical Apply no Standby:

```
ALTER DATABASE START LOGICAL STANDBY APPLY IMMEDIATE;
```

- Habilitar a configuração do Data Guard Broker:

```
dgmgrl /
DGMGRL> ENABLE CONFIGURATION;
DGMGRL> exit
```

- Ajustes do ambiente CARGO_ODS. Devido a problemas para aplicar as alterações nas tabelas de interface elas foram removidas da replicação.

```
alter database stop logical standby apply;
EXECUTE DBMS_LOGSTDBY.SKIP (stmt => 'DML', schema_name => 'CARGO',
                      object_name => 'SYNITF_DOF', proc_name => NULL);
EXECUTE DBMS_LOGSTDBY.SKIP (stmt => 'DML', schema_name => 'CARGO',
                      object_name => 'SYNITF_IDF', proc_name => NULL);
ALTER DATABASE START LOGICAL STANDBY APPLY IMMEDIATE;
```

Configuração do Data Guard Broker

Habilitar o inicio automático do Data Guard Broker, deve ser realizado tanto no banco primário como no secundário.

```
SQL> alter system set dg_broker_start=TRUE scope=both sid='*';
```

Configurando o Data Guard:

```
$ export ORACLE_SID=<SID>
$ dgmglr /
```

Criando a configuração:

```
DGMGRL> CREATE CONFIGURATION <config_name> AS PRIMARY DATABASE IS <database_name> CONNECT IDENTI
```

Adicionando o banco standby:

```
DGMGRL> ADD DATABASE <standby_database_name> as CONNECT IDENTIFIER IS <standby_tnsnames> MAINTAI
```

Habilitando o Data Guard:

```
DGMGRL> enable configuration;
```